

# AI DAY NOIDA

Detailed Problem Statements

Powered by Nerds Room

20 September 2026 • Paytm Office, Noida

AI Buildathon & Demo Day

Build. Demo. Showcase.

This document defines four challenge tracks designed to guide participants toward practical, AI-enabled solutions while leaving room for different technical approaches.

# PROBLEM STATEMENT 01

## Community Space — AI-Powered Community Intelligence Platform

### Background

Community groups, colleges, developer communities and event organizers often depend on scattered messaging groups, social media, spreadsheets and forms. This makes it difficult to discover relevant events, find people with complementary skills, manage knowledge and maintain useful community information.

### Challenge

Build an AI-powered community platform that helps people discover relevant communities, events, resources and collaborators while helping organizers manage community knowledge and engagement.

### Core Requirements

- Create searchable profiles containing skills, interests and areas of expertise.
- Recommend relevant events, communities, resources or collaborators based on user intent.
- Allow organizers to publish events, announcements and resources in a structured format.
- Use AI to summarize long announcements, event descriptions or community discussions.
- Provide semantic search so users can ask natural-language questions about community information.
- Support intelligent teammate/collaborator matching based on skills and project requirements.
- Include moderation or reporting mechanisms for misleading, duplicate or inappropriate content.

### Advanced Directions

- RAG-based community knowledge assistant.
- Semantic recommendation and matching.
- AI-generated event/resource summaries.
- Conversation-based discovery interface.
- Organizer analytics showing engagement and frequently requested information.

# PROBLEM STATEMENT 02

## Artificial Intelligence — AI Agent for Real-World Productivity

### Background

Many users spend significant time converting unstructured information into actions. Emails, documents, notices, project requirements and meeting notes can contain important information but require manual reading, prioritization and task creation.

### Challenge

Build an AI agent or intelligent assistant that can understand user-provided information, reason over it and execute or recommend useful actions through controlled workflows.

### Core Requirements

- Ingest documents, notes, text or other permitted sources.
- Extract tasks, deadlines, entities, priorities and decisions.
- Answer questions using grounded information from the user's knowledge base.
- Generate actionable plans rather than only summaries.
- Use tools or APIs where appropriate to complete defined tasks.
- Show sources, reasoning context or confidence indicators where practical.
- Handle uncertainty and clearly state when information is unavailable.

### Advanced Directions

- Agentic workflows with tool calling.
- RAG and vector search.
- Structured extraction and classification.
- Multi-step task planning.
- Human-in-the-loop approval for sensitive actions.
- Evaluation using task completion, grounding or retrieval metrics.

# PROBLEM STATEMENT 03

## AI for Sustainability — Intelligent Resource Optimization

### Background

Energy, water, waste and other resources are frequently consumed without detailed visibility into where or why waste occurs. A useful AI system can identify patterns, detect anomalies and recommend interventions.

### Challenge

Create an AI-powered system that monitors, predicts or optimizes resource usage for a campus, office, building, residential environment or community.

### Core Requirements

- Use real or simulated resource-consumption data.
- Detect unusual consumption or potential wastage.
- Forecast future usage using historical patterns.
- Generate recommendations to reduce consumption.
- Estimate potential cost, energy or environmental savings.
- Present trends and alerts through an understandable dashboard.
- Explain why a recommendation or anomaly was generated.

### Advanced Directions

- Time-series forecasting.
- Anomaly detection.
- Computer vision for waste classification.
- Optimization and recommendation models.
- IoT/sensor integration or realistic simulated sensor data.

# PROBLEM STATEMENT 04

## Open Innovation — Build What Matters with AI

### Background

Open Innovation gives participants the freedom to identify a meaningful real-world problem instead of solving a predefined use case. The solution may target education, accessibility, finance, healthcare support, agriculture, productivity, cybersecurity, developer tools, campus life or another domain.

### Challenge

Identify a real problem, validate its importance, design an AI-enabled solution and demonstrate a working prototype. The strongest submissions should show a clear connection between the problem and the AI capability being used.

### Core Requirements

- Clearly define the target user and pain point.
- Explain existing alternatives and the gap being addressed.
- Demonstrate a working core user flow.
- Use AI where it provides a meaningful advantage.
- Define at least one measurable success or impact metric.
- Explain data requirements, model/API dependencies and limitations.
- Describe how the prototype could be scaled or deployed in the real world.

### Advanced Directions

- Accessibility and assistive AI.
- Education and personalized learning.
- Healthcare-support workflows.
- Financial awareness and productivity.
- AI developer tools.
- Cybersecurity and privacy.
- Creative AI and multimodal experiences.
- Agentic automation and intelligent workflows.

# Submission & Evaluation Guidance

Every submission should make it easy for the jury to understand the problem, the AI contribution and the working product.

## Minimum Submission Expectations

- Working prototype with a clear end-to-end demonstration.
- Problem statement and target-user definition.
- Architecture/technology overview.
- Clear explanation of the AI model, agent, retrieval or ML component.
- Sample data or demonstration dataset where required.
- Known limitations and future roadmap.

## Evaluation Dimensions

- Problem relevance and clarity.
- Quality and meaningfulness of AI implementation.
- Technical depth and execution.
- Innovation and originality.
- User experience and usability.
- Practical feasibility and scalability.
- Demonstrated impact.
- Quality of final demo and explanation.

## Important Note

Participants are encouraged to choose the challenge that best matches their interests and capabilities. The descriptions define the problem space, not a single prescribed implementation. Teams may use different models, frameworks, APIs and architectures as long as the resulting solution addresses the stated challenge and can be demonstrated responsibly.

AI DAY NOIDA • POWERED BY NERDS ROOM